

ISOMETRIC

SIDE

TOP

CUTAWAY — internal drive revealed

KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius150 mm (Ø300)

ball/params.py bludger.radius = 0.15 m

Mass1.2 kg (foam)

bludger.mass – looks iron, isn't

Top speed15 m/s

bludger.max_speed

Max accel22 m/s²

bludger.max_accel (most aggressive)

Lift4 × EDF (internal)

swarm core

Soft-tag radius1.5 m

tag_radius (outside no-contact floor)

Bat deflect1.6 m

bat_radius

Contact0 — software hit

ball/avoidance.py, worst clr +0.40 m

NOTE

All threat, no contact. The 'hit' is a proximity tag registered in software ~1.5 m out; the foam shell never has to reach a person.

BILL OF MATERIALS — BLG (12 line items, prototype qty 1)

REF	PART	SPEC	QTY	\$ EA	\$ EXT
BLG-01	Black impact-foam sphere shell	Ø300 mm (R=150 mm), riveted-iron look	1	70	70
BLG-02	Equator rivet bands (cosmetic)	two crossed bands + dummy rivets	1	18	18
BLG-03	Ducted-fan lift unit	Ø40 mm internal, internal swarm	4	40	160
BLG-04	Brushless motor	per fan, high power-to-weight	4	22	88
BLG-05	4-in-1 ESC	20 A, telemetry	1	38	38
BLG-06	Flight controller	IMU on-board, lead-pursuit guidance	1	42	42
BLG-07	LiDAR / proximity sensor	target ranging for the soft tag	2	55	110
BLG-08	Bat-swing detector	sees the beater's marked bat	1	16	16
BLG-09	No-contact avoidance compute	clamps velocity to a stop at the shell	1	0	0
BLG-10	Li-Po pack	3S, sized for sprint pursuits	1	24	24
BLG-11	Status LED	angry red glow	1	5	5
BLG-12	Telemetry link	reports pose to the Dementor	1	14	14

UNIT TOTAL (prototype):\$585